# Capstone Train, Analyze, Improve

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

## One complete comparison and freeze record.

Name fixed factors,trace all gains and misses,and distinguish validation from final evidence.

**Keep one habit:** Fresh W1–W3/W5 → selected separate frozen matrix → project freeze → changed-case return.

MAIN LESSON · SECTION 1

## Recall · Retrieve before notes

> Keep this task's complete inputs, stated rule and support record together.

### Task and evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Capstone Dataset and Baseline. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Proxy Features and Fairness. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Feature Engineering. Give an example and a mistake that the idea helps you avoid.

### Capstone Dataset and Baseline

**My remembered idea, example and correction**

### Proxy Features and Fairness

**My remembered idea, example and correction**

### Feature Engineering

**My remembered idea, example and correction**

# W1 · State the controlled fit

> Keep the complete current fixture beside a fresh response;earlier final-matrix practice stays as review.

## Complete kit reference

Complete72-row parcel kit:40 training, 20 validation, 12 reserved final feature rows with blank references. Baselineclear fitted from 22 clear/18 blocked training labels. V1 width-only, V2 width+height; training-only lows/ranges 2/8 and 2/7, k1, Manhattan, exact-distance ID ties. Validation TN/FP/FN/TP baseline 9/0/11/0, V1 9/0/7/4, V2 9/0/3/8; correct 9/13/17of20.

V2 gainsP42/P49/P60/P67, no regressions; missesP24/P56/P64. Choose most validation-correct, then fewer features, then name:V2. Save freeze before teacher reference release. No actual kit final score is supplied to students. Selected independent matrix is a separate exercise.

## Current task and response

- Name V1, V2, the single change and the fixed training/validation/learner/preprocessing/tie rules.

**My capstone comparison, freeze and evidence reasoning**

<details>
<summary>▼ Review · Earlier matrix task and saved response</summary>

Reference for this exercise: open this week's student lesson, "Capstone Train, Analyze, Improve", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

Compute the worked candidate metrics and baseline accuracy.
</details>

**My reasoning and evidence**

MAIN LESSON · SECTION 13

# W2 · Trace the full comparison

> Keep the complete current fixture beside a fresh response;earlier final-matrix practice stays as review.

## Complete kit reference

Complete72-row parcel kit:40 training, 20 validation, 12 reserved final feature rows with blank references. Baselineclear fitted from 22 clear/18 blocked training labels. V1 width-only, V2 width+height; training-only lows/ranges 2/8 and 2/7, k1, Manhattan, exact-distance ID ties. Validation TN/FP/FN/TP baseline 9/0/11/0, V1 9/0/7/4, V2 9/0/3/8; correct 9/13/17of20.

V2 gainsP42/P49/P60/P67, no regressions; missesP24/P56/P64. Choose most validation-correct, then fewer features, then name:V2. Save freeze before teacher reference release. No actual kit final score is supplied to students. Selected independent matrix is a separate exercise.

## Current task and response

- Using the complete parcel kit, explain the fixed learner and the single V1-to-V2 feature change. Reconstruct the validation matrices and all gained/regressed case IDs. Choose using the predeclared rule, then state what must be saved before final labels are released.

**My capstone comparison, freeze and evidence reasoning**

▼ **Review · Earlier matrix task and saved response**

What does the one-correct-case improvement establish?

**My reasoning and evidence**

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MAIN LESSON · SECTION 14

# W3 · Repair a test-status claim

> Keep the complete current fixture beside a fresh response;earlier final-matrix practice stays as review.

## Complete kit reference

Complete72-row parcel kit:40 training, 20 validation, 12 reserved final feature rows with blank references. Baselineclear fitted from 22 clear/18 blocked training labels. V1 width-only, V2 width+height; training-only lows/ranges 2/8 and 2/7, k1, Manhattan, exact-distance ID ties. Validation TN/FP/FN/TP baseline 9/0/11/0, V1 9/0/7/4, V2 9/0/3/8; correct 9/13/17of20.

V2 gainsP42/P49/P60/P67, no regressions; missesP24/P56/P64. Choose most validation-correct, then fewer features, then name:V2. Save freeze before teacher reference release. No actual kit final score is supplied to students. Selected independent matrix is a separate exercise.

## Current task and response

- V2 is selected using 17/20 validation and its three mistakes. Can the same 20 rows now be described as an untouched final test for V2? Explain the evidence status.

**My capstone comparison, freeze and evidence reasoning**

▼ **Review · Earlier matrix task and saved response**

A threshold changes after test errors are read. Label its score on the same rows.

**My reasoning and evidence**

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MAIN LESSON · SECTION 16

# W5 · Freeze and plan final reporting

> Keep the complete current fixture beside a fresh response;earlier final-matrix practice stays as review.

## Complete kit reference

Complete72-row parcel kit:40 training, 20 validation, 12 reserved final feature rows with blank references. Baselineclear fitted from 22 clear/18 blocked training labels. V1 width-only, V2 width+height; training-only lows/ranges 2/8 and 2/7, k1, Manhattan, exact-distance ID ties.

Validation TN/FP/FN/TP baseline 9/0/11/0, V1 9/0/7/4, V2 9/0/3/8; correct 9/13/17of20.

V2 gainsP42/P49/P60/P67, no regressions; missesP24/P56/P64. Choose most validation-correct, then fewer features, then name:V2. Save freeze before teacher reference release. No actual kit final score is supplied to students. Selected independent matrix is a separate exercise.

### Current task and response

- Write the freeze fields and teacher-release gate. Explain how final baseline/candidate results and any later revised version must be labelled without inventing a final score.

**My capstone comparison, freeze and evidence reasoning**

▼ **Review · Earlier matrix task and saved response**

What must remain visible when version 2 is proposed without a fresh test?

**My reasoning and evidence**

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MAIN LESSON · SECTION 4

## Try alone · Selected experiment task

> Keep this task's complete inputs, stated rule and support record together.

### Task and evidence record

- This is a separate fully supplied 12-case frozen-matrix exercise, not the parcel kit's hidden result. Rows are true clear/blocked and columns predicted clear/blocked; blocked is positive. Use its given counts and compare both kinds of error. Keep the first attempt before opening the teacher key; do not substitute the worked validation or pictured 16-case numbers.

- Frozen candidate[[6, 2],[1, 3]] and baseline[[8, 0],[4, 0]] on 12 cases. Compute candidate metrics and compare error types.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

#### Independent case: Apply independently

> **My independent explanation, trace and limits**

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MAIN LESSON · SECTION 9

## Project · Keep the experiment evidence

> Keep this task's complete inputs, stated rule and support record together.

### Task and evidence record

- Finalize the Plant Classifier experiment record, confusion matrix, subgroup table, model card, and honest remaining-limit statement.

- The project change and **evidence** I will keep

**Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

# Feedback · Repair and return

> Keep this task's complete inputs, stated rule and support record together.

## Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

| Evidence | My record |
|---|---|
| Worked alone / hint / model | |
| First step I repaired and why | |
| New case checked later; date and result | |
| What I can now explain without notes | |
| One remaining question | |

MAIN LESSON · SECTION 10

# Finish · Judge your evidence

> Keep this task's complete inputs, stated rule and support record together.

## Task and evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.
  - I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

# Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

## Reference · Supported W2 record

> Keep this task’s complete inputs, stated rule and support record together.

### Task and evidence record

- Earlier final-matrix W2 record:retain its saved response as review. Use current W2for the complete kit training,validation and freeze comparison.

| Given inputs | Procedure I chose | Middle steps | Result and check |
|---|---|---|---|
| | | | |

**The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

## Extra practice · Individual reasoning check

> Keep this task’s complete inputs, stated rule and support record together.

### Task and evidence record

- Use the worked case for Capstone evaluation and revision. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier learning target

REFERENCE · SECTION 5

## Reference · Earlier learning target

> Keep this task’s complete inputs, stated rule and support record together.

### Task and evidence record

- A new model reduces validation errors from 8 to 5 but the team also removed three hard cases. Can improvement be attributed to the model? Propose a repair to the experiment.

▼ **Extra practice · Earlier complete case**

REFERENCE · SECTION 6

## Extra practice · Earlier complete case

> Keep this task's complete inputs, stated rule and support record together.

### Task and evidence record

- Baseline validation accuracy**70%**; model**80%**; most remaining errors are dim images.
- Plan improvement **evidence**.
- Show your trace, calculation or **evidence**

**Write or draw your reasoning and evidence.**

▼ Extra practice · Changed case

REFERENCE · SECTION 7

## Extra practice · Changed case

> Keep this task's complete inputs, stated rule and support record together.

### Task and evidence record

- A revision raises overall validation accuracy from **80%** to 84% but lowers dim-light accuracy from **60%** to 40%. How should the improvement claim change?

- My independent answer and reason

> **Write or draw your reasoning and evidence.**

**Week 35 Test and reflect**

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

## Extra practice · Design a revealing test

> Keep this task's complete inputs, stated rule and support record together.

### Task and evidence record

- Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

| Test record | My evidence |
| --- | --- |
| Changed input or condition | |
| Expected answer or fallback | |
| Observed result | |
| Why the result occurred | |

▼ **Extra practice · Earlier transfer**

REFERENCE · SECTION 11

## Extra practice · Earlier transfer

> Keep this task's complete inputs, stated rule and support record together.

### Task and evidence record

- A new model reduces validation errors from 8 to 5 but the team also removed three hard cases. Can improvement be attributed to the model? Propose a repair to the experiment.

- Record support used: none, hint or teacher explanation.

**My independently reasoned answer**

▼ **Reference · Earlier W4 record**

REFERENCE · SECTION 15

# Reference · Earlier W4 record

> Keep this task's complete inputs, stated rule and support record together.

## Task and evidence record

- Matrix convention: rows are reference [clear, blocked]; columns are prediction [clear, blocked]; blocked is positive. Each matrix is [[TN, FP], [FN, TP]]. Use the same twelve reference cases for candidate and baseline.

- Frozen candidate[[6, 2],[1, 3]] and baseline[[8, 0],[4, 0]] on 12 cases. Compute candidate metrics and compare error types.

**My reasoning and evidence**

▼ Extra practice · Pictured case

REFERENCE · SECTION 18

# Extra practice · Pictured case

> Keep this task's complete inputs, stated rule and support record together.

## Task and evidence record

- Choose the next capstone revision from explicit false alarms and misses.
- Positive=spotted. CandidateTP4, FN 2, FP 1, TN 9; baselineTP0, FN 6, FP 0, TN 10. Both evaluate the same16 cases.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- Why is "the revision is better" incomplete? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-4/35

# Candidate, before revision

| | predicted spotted | predicted healthy |
|---|---|---|
| actual spotted | 4 | 2 |
| actual healthy | 1 | 9 |

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/35

# Capstone Train, Analyze, Improve

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.

### 1. Compare accuracy

Candidate13/16=81.25%; baseline10/16=62.5%.

### 2. Compare detection

Candidate catches4/6 spotted leaves; baseline catches0/6.

### 3. Inspect errors

Review candidate's two misses and one false alarm with conditions and label evidence before revising.

**CHANGE ONE THING**

A revision fixes one miss but creates two extra false alarms.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

**My picture-based explanation and changed-case prediction**

REFERENCE · SECTION 19

# Reference · Project portfolio milestone

> Keep this task's complete inputs, stated rule and support record together.

## Task and evidence record

- Project: Responsible Plant Classifier
- Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.
- For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

| Evidence to keep | My record |
| --- | --- |
| This week's input and deciding step | |
| Prediction and actual result (or labelled paper trace) | |
| One change since the previous checkpoint and why | |
| One error or limit and the next test | |

**Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 20

# Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

## First supported question and hint

- Choose the next capstone revision from explicit false alarms and misses.
- Which labelled step matches this detail? Review candidate's two misses and one false alarm with conditions and label **evidence** before revising.
- 1. Inspect errors
- 2. Compare accuracy
- 3. Compare detection
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Practice 1: explain the deciding evidence

**My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Inspect errors Review candidate's two misses and one false alarm with conditions and label **evidence** before revising. Read the picture in order.

Review candidate's two misses and one false alarm with conditions and label **evidence** before revising. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

## Changed-case question and hint

- Changed-case practice: A revision fixes one miss but creates two extra false alarms. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. TP 5, FN 1, FP 3, TN 7 gives 12/16=75% accuracy and 5/6 recall.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Practice 2: explain the deciding evidence

**My selected conclusion, deciding fact and explanation**

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: TP 5, FN 1, FP 3, TN 7 gives 12/16=75% accuracy and 5/6 recall. Recall rises while accuracy falls; the preferred revision depends on declared error consequences.

## Repair and keep the record

Review Capstone Dataset and Baseline (ai-4/34). Goal: Define a feasible capstone question and acceptance criteria. Short practice: What provenance field prevents the crop from appearing independent?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Proxy Features and Fairness (ai-4/32). Goal: Identify possible proxy features. Short practice: What **evidence** contradicts the original “all setting information removed” claim?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Feature Engineering (ai-4/29). Goal: Create features available at **prediction** time. Short practice: What should be saved so another group transforms the query identically?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

**What I repaired, the evidence I kept and the next test**

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

**Repair a missing building block**

**Keep your completed work**

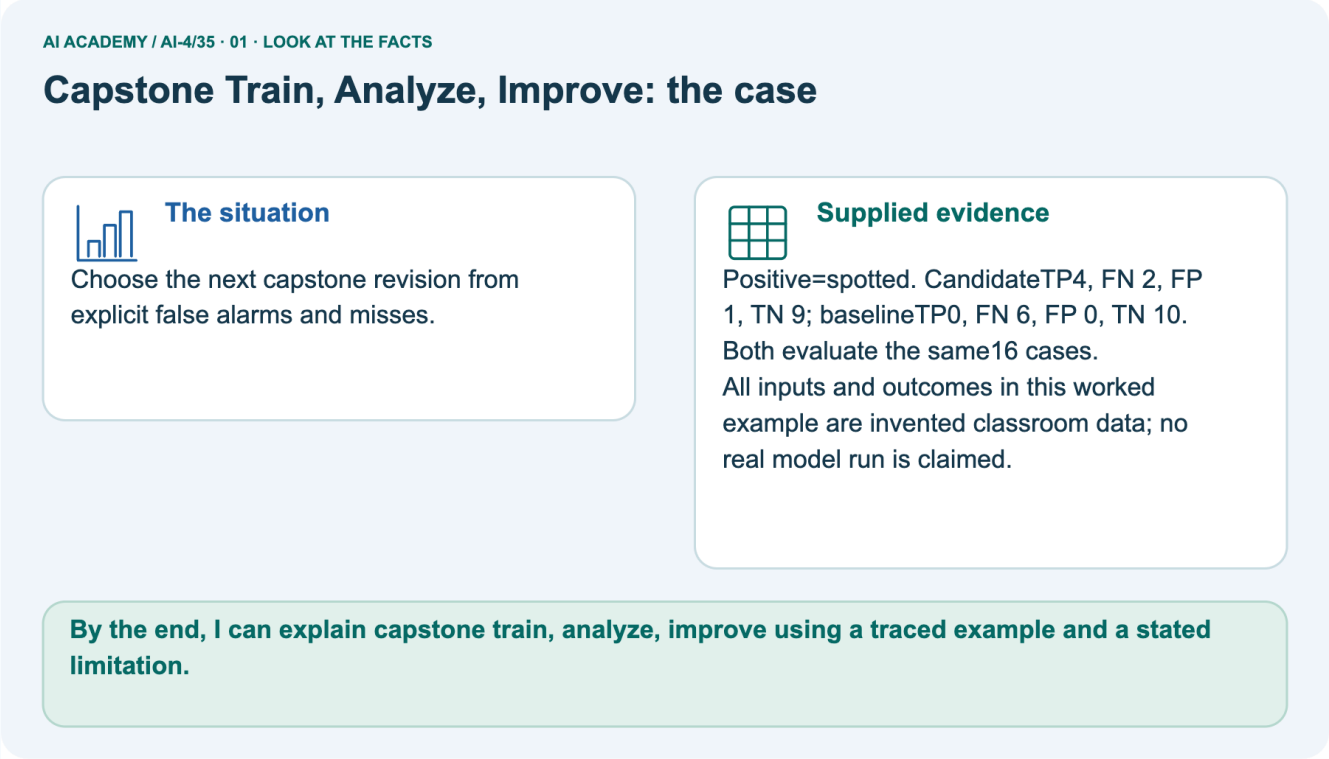
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

## Reference · Six-picture response sheets

> Choose the relevant picture instead of repeating every task.

## 01. Capstone Train, Analyze, Improve: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

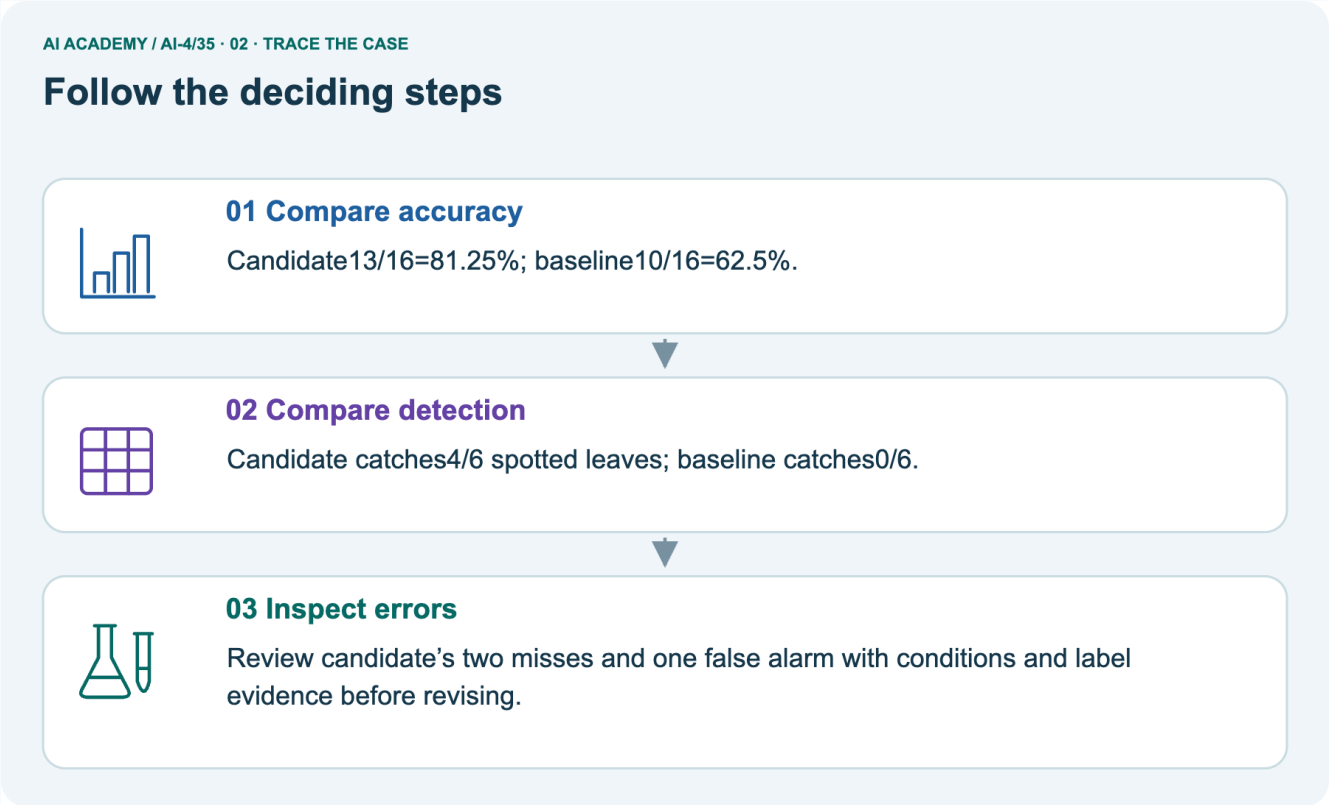
Choose the next capstone revision from explicit false alarms and misses. Supplied facts: Positive=spotted. CandidateTP4, FN 2, FP 1, TN 9; baselineTP0, FN 6, FP 0, TN 10. Both evaluate the same16 cases.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

### My picture explanation, deciding evidence and one limit

## 02. Follow the deciding steps



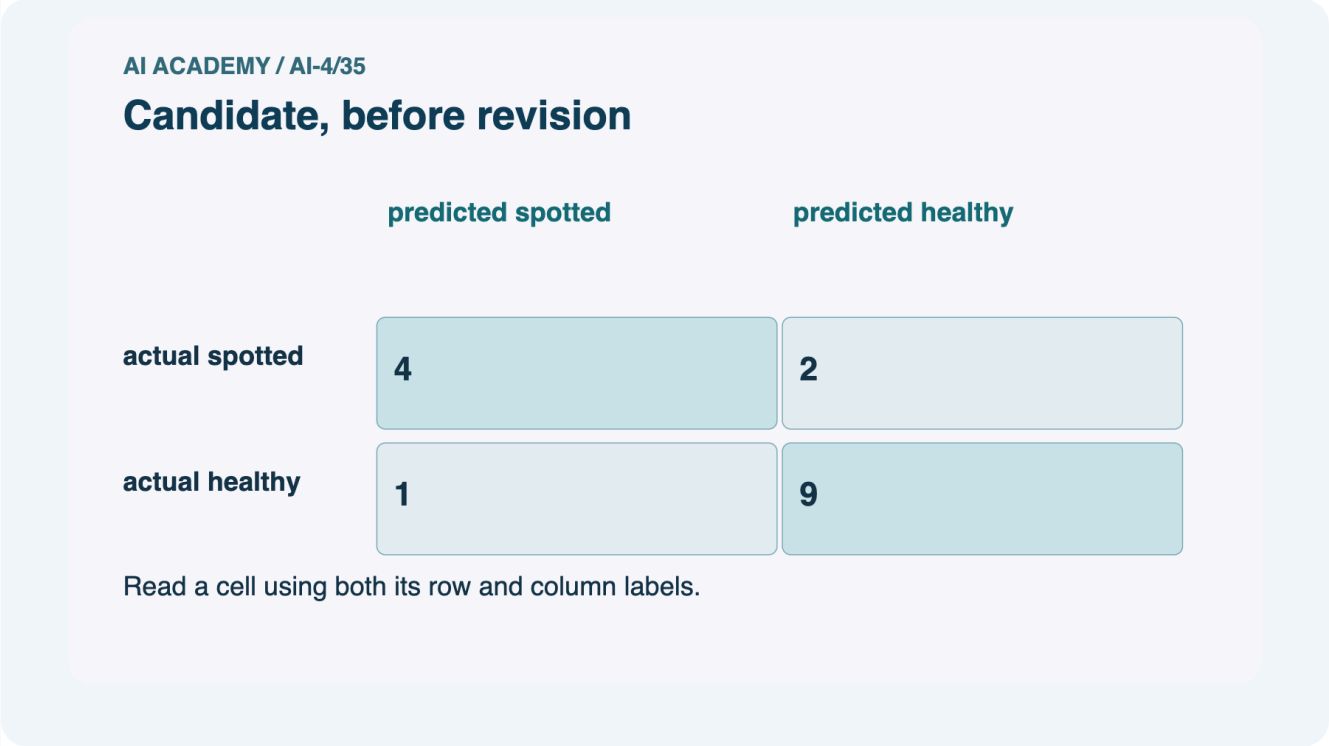

Follow the labelled picture with the matching explanation. Open the vector diagram

Compare accuracy: Candidate13/16=81.25%; baseline10/16=62.5%. → Compare detection: Candidate catches4/6 spotted leaves; baseline catches0/6. → Inspect errors: Review candidate's two misses and one false alarm with conditions and label **evidence** before revising.

Point to each numbered stage and explain which recorded **input** it uses.

### My picture explanation, deciding evidence and one limit

## 03. Candidate, before revision



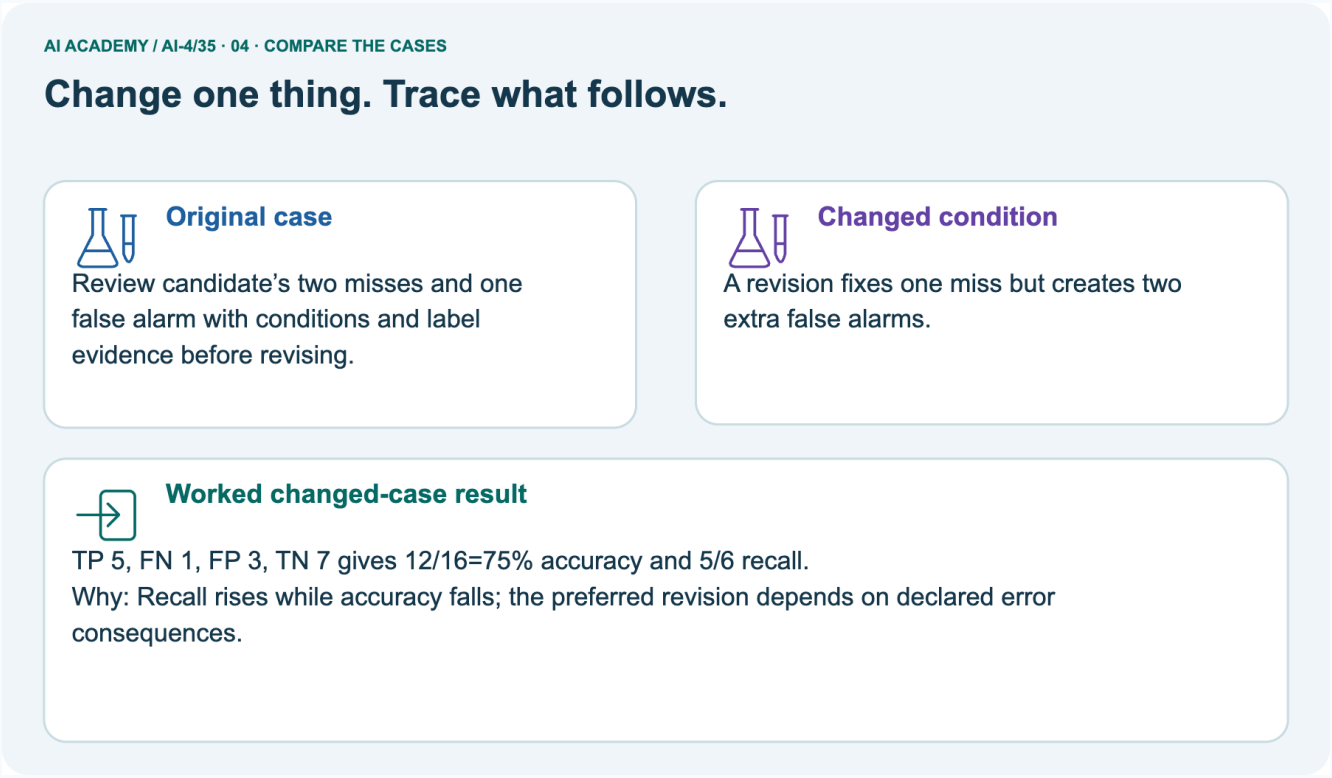

Follow the labelled picture with the matching explanation. Open the vector diagram

Candidate, before revision. actual spotted: predicted spotted = 4, predicted healthy = 2. actual healthy: predicted spotted = 1, predicted healthy = 9.

Read the labels and values. Explain the relationship the model shows.

### My picture explanation, deciding evidence and one limit

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

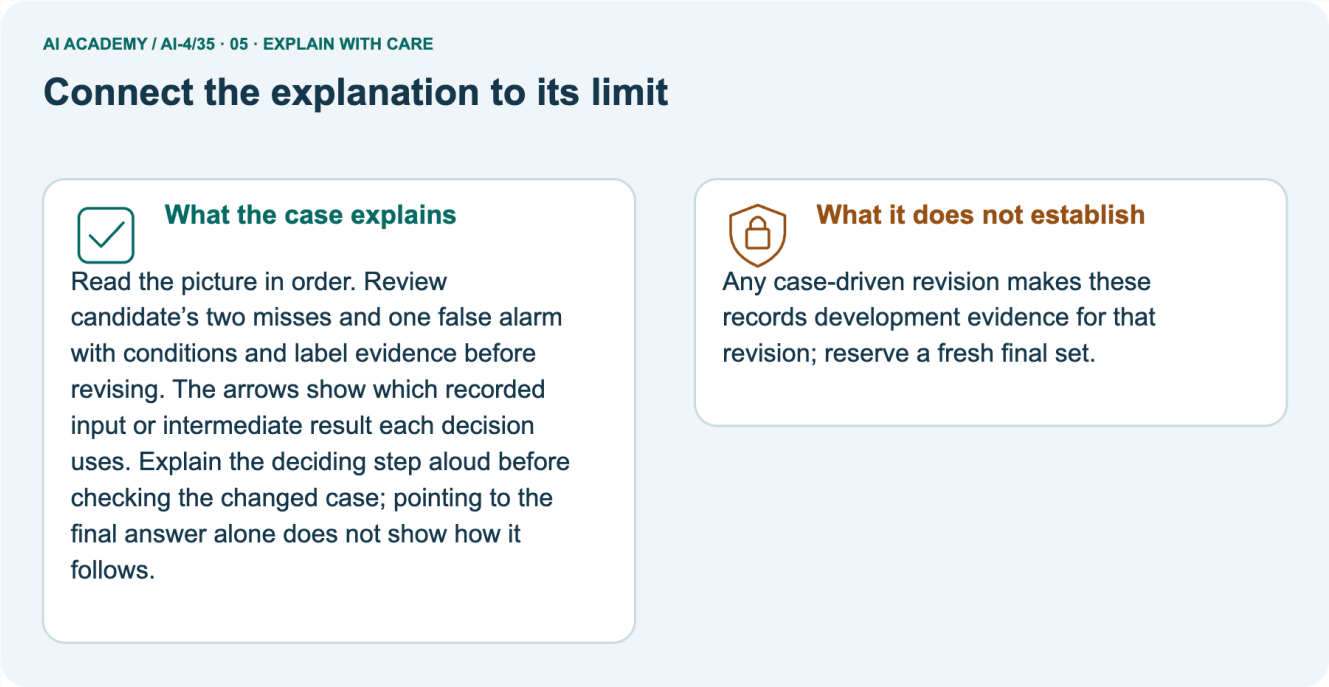
Original case: Review candidate's two misses and one false alarm with conditions and label **evidence** before revising. Changed condition: A revision fixes one miss but creates two extra false alarms.

Expected result: TP 5, FN 1, FP 3, TN 7 gives 12/16=75% accuracy and 5/6 recall. Reason: Recall rises while accuracy falls; the preferred revision depends on declared error consequences.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

### My picture explanation, deciding evidence and one limit

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Review candidate's two misses and one false alarm with conditions and label **evidence** before revising. The arrows show which recorded **input** or intermediate result each decision uses.

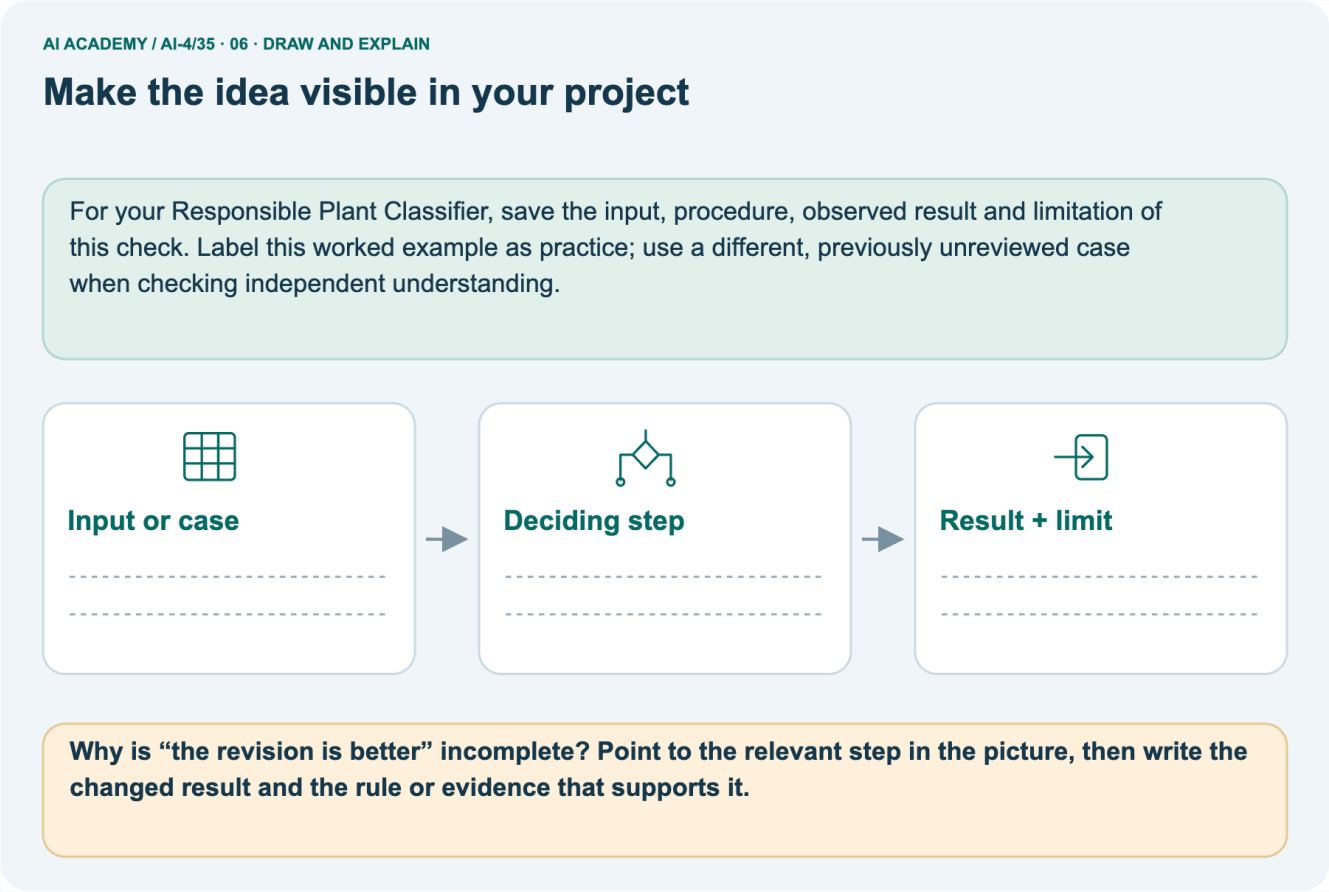
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Any case-driven revision makes these records development **evidence** for that revision; reserve a fresh final set.

Explain the case, then point to a limit on the claim.

### My picture explanation, deciding evidence and one limit

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why is “the revision is better” incomplete? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

**My picture explanation, deciding evidence and one limit**

▼ Earlier wording, alternatives and picture transcripts

**01. Capstone Train, Analyze, Improve: the case**

**02. Follow the deciding steps**

**03. Candidate, before revision**

**04. Change one thing. Trace what follows.**

**05. Connect the explanation to its limit**

**06. Make the idea visible in your project**

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.